

Nallabati Venkata Durga Sai Sarat Chandra

Target Role: Embedded Software & Firmware Intern | Texas Instruments

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SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive software architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech) , Amrita Vishwa Vidyapeetham CGPA: 9.05 / 10.00	2024 – Present Amritapuri
Higher Secondary Education , Sri Chaitanya Vidhyanikethan CBSE (MPCEA) Percentage: 88.2%	2024 Vishakapatnam
Secondary School Education , Sri Chaitanya School SSC AP Marks: 570 / 600	2022 Rajam

SKILLS

Firmware & Languages: Embedded C (C99/C11), Modern C++, FreeRTOS, Bare-Metal ARM Cortex-M, Bash, POSIX Sockets
Silicon & Peripherals: STM32 (ARM Cortex-M), ESP32, CAN Bus (2.0B), DMA Circular UART (921,600 baud), I2C (9-Clock Recovery), SPI
IoT, Telemetry & Hardware: Valetron A7670C (4G LTE Cat-1), Opto-Isolated 240V Relays, RC Snubbers, MAX30102 PPG, Watchdogs (IWDG)
Instrumentation & Tooling: Digital Storage Oscilloscopes, Logic Analyzers, KiCAD, Linux, PlatformIO, Git/GitHub, Tectonic LaTeX

PROJECTS

Industrial GSM & 4G LTE Cellular IoT Gateway with Dual-Mode Failover Valetron A7670C 4G LTE Cat-1, ESP32, FreeRTOS, Circular DMA UART, 240V Relays • <1.8s automatic network failover (WiFi <-> 4G LTE); opto-isolated 240V relay switching with RC snubbers.	Embedded Firmware Engineer
Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS • <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.	Biomedical Systems Developer
Pasupatha Autonomous Companion Robot Control Bus ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed • 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).	Head of Embedded Systems, Club Mirai

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity , IEEE ICRM 2025 • Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.	11/2025
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ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.